

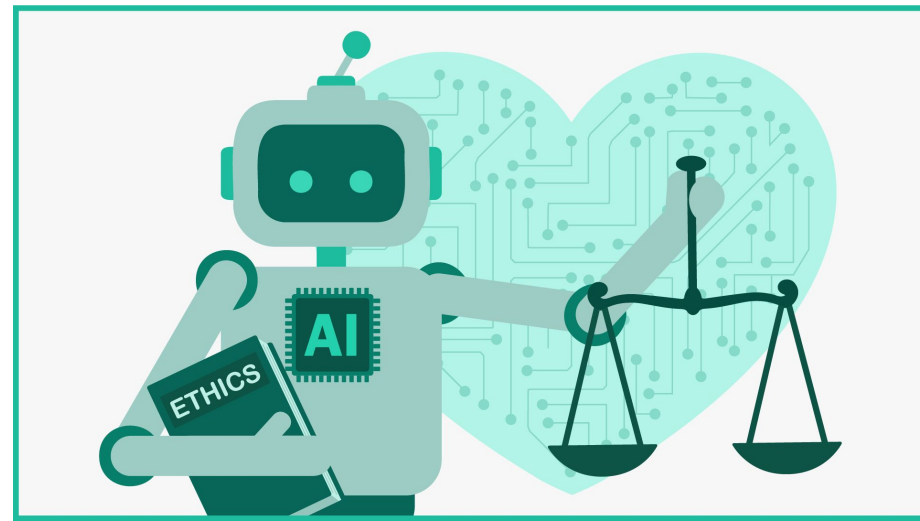
AI Ethics — Responsible AI

Understanding the moral, professional, and social boundaries of Artificial Intelligence

By Divesh Jadhvani

Introduction — Ethics Overview

- Ethics: Line between what is right and wrong
- Professional Ethics
- Internet Ethics
- Research Ethics
- AI Ethics



What is AI Ethics?

- “Ethics” means a set of moral principles about what is right or wrong.
- AI ethics refers to the field that studies how we should design, build, deploy and use AI systems so they abide by human values, minimize harm and maximize benefit.

Case Studies Overview

- 1. Amazon Screening Tool
- 2. Face Recognition System
- 3. UK Work Pension Benefit
- 4. Criminal Justice (COMPAS)
- 5. Healthcare AI Failure
- 6. interview Cracking software
- (linkedin)



Case Study 1 — Amazon Screening Tool

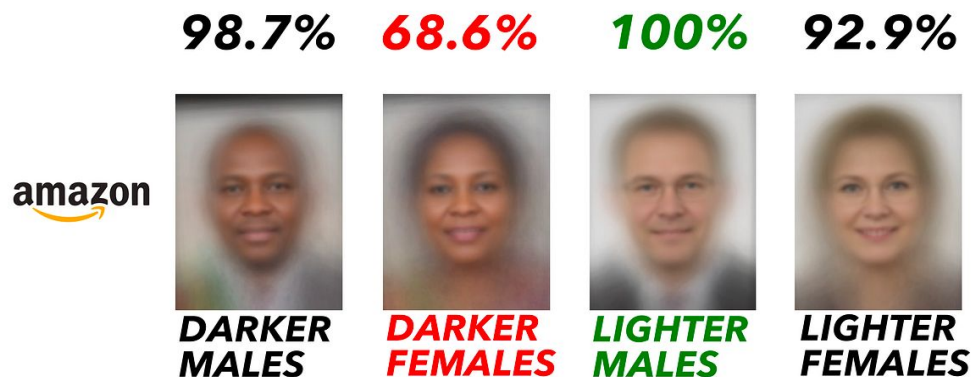
- AI favored male applicants due to biased training data.
- Lesson: Bias detection and fairness testing are essential.



Case Study 2 — Face Recognition Bias

- Higher error rates for darker-skinned women (MIT study).
- Lesson: Diversity in training data ensures fairness.

August 2018 Accuracy on Facial Analysis Pilot Parliaments Benchmark



Amazon Rekognition Performance on Gender Classification

Case Study 3 — UK Work Pension Benefit

- AI wrongly flagged legitimate claims as fraud.
- Lesson: Human oversight and transparency are vital.



Case Study 4 — Criminal Justice (COMPAS)

- Black defendants misclassified as high risk nearly 2× more.
- Lesson: Fairness and accountability critical in justice systems.



Case Study 5 — Healthcare AI

- AI sepsis detection tool overstated accuracy.
- Lesson: Transparency and validation save lives.



Why Study AI Ethics?

- AI impacts health, hiring, finance, and law enforcement.
- Without ethics → bias, harm, loss of trust, privacy invasion.
- Ethical AI builds trust and social acceptance.

What Happens If We Don't Have AI Ethics

- Biased algorithms and unfair outcomes
- Privacy violations and misuse
- Loss of public trust
- Regulation backlash slowing innovation

Pillars of AI Ethics

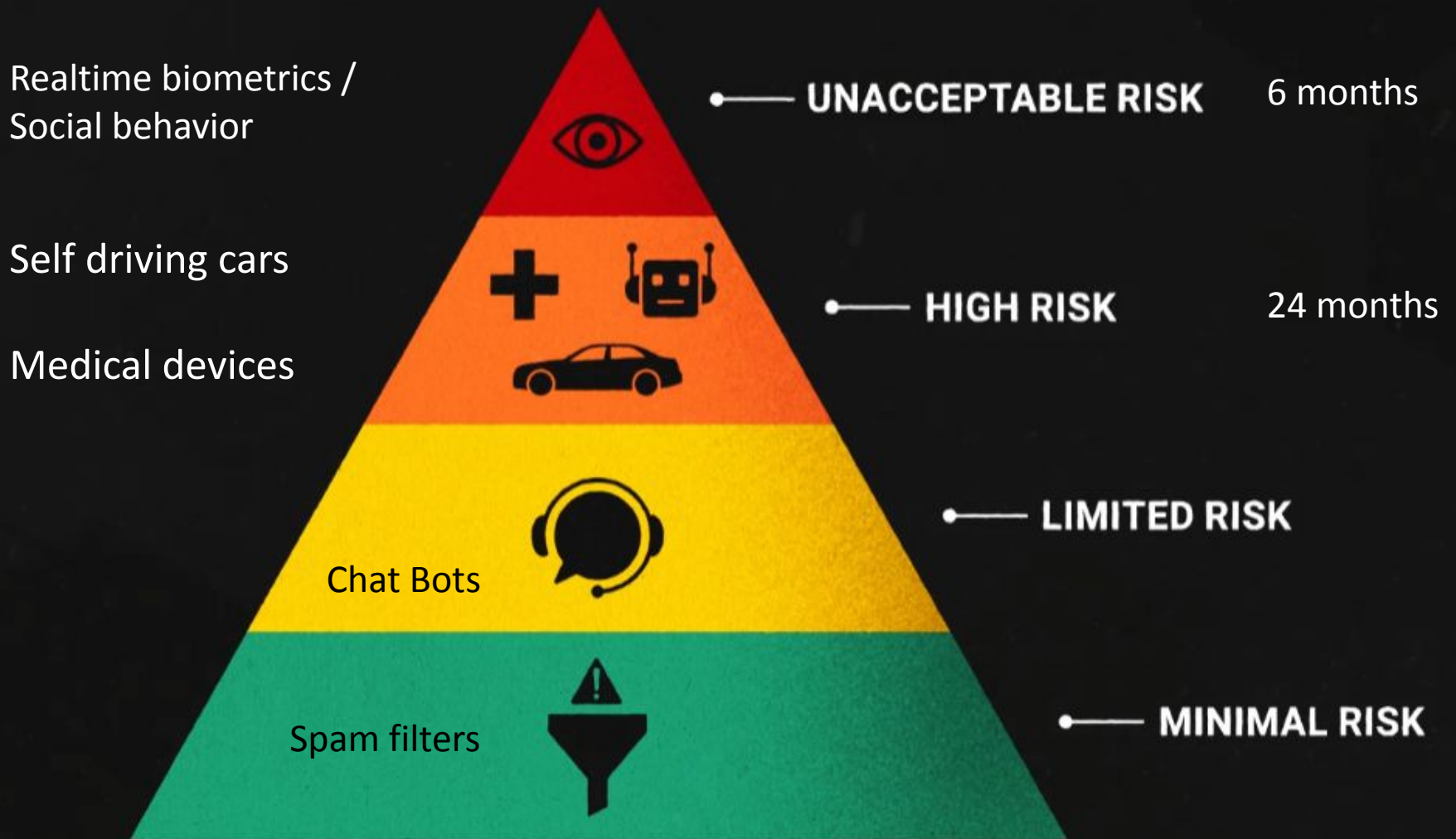
- 1. Fairness – same rule for all groups
- 2. Transparency - Explainability
- 3. Accountability — define system limits before design
- 4. Privacy — encrypt user data
- 5. Security & Robustness — prevent prompt injection, password leaks
- 6. Beneficence — help, not replace humans

Bias Detection & Fairness Metrics

- Bias Detection: Identifying unfairness in data or outcomes.
- Fairness Metrics: Tools to measure equality (e.g., demographic parity, equal opportunity).
- Eg : loan approval
- If 70 % men get approval , 70 % females should also get approval

AI Regulation

- Laws and policies for safe AI use.
- Ensures accountability, transparency, and rights protection.
- Example: EU AI Act — classifies AI systems by risk levels.



GDPR = Your Data, Your Rights

General Data Protection Regulation

EU customer rights below

- Gives consumers rights to launch complaints about AI systems
- Fines businesses that fail to comply with regulations
- Establishes transparency requirements for AI systems

Google

50 million

for lack of transparency
and validity consent

BRITISH AIRWAYS

20 million

for a data breach affecting
over 400,000 customers

GDPR :

Data subject
Data controller
Data processor
Personal data

Rights to users :

Right to inform
Right to access
Right to Rectification
Right to erase – if ac
etc

[REF](#)

Summary & Reflection

- Ethics helps align AI with humanity.
- Bias detection → Fairness → Regulation → Trustworthy AI.
- “Technology defines capability; Ethics defines direction.”